Project: Worked examples
Title: Fault protection: Zs, disconnection, adiabatic, fault level
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1. Scope and method

Automatic disconnection of supply in a TN system for one circuit: the earth fault loop impedance at the conductor's operating temperature against the device's maximum, the disconnection time, the protective conductor (and line conductor) against the adiabatic minimum, and the prospective three-phase fault at the origin against the device's breaking capacity.

WHAT IT CALCULATES	RELATION
Loop impedance	$Z_s = Z_e + (R_1 + R_2)\cdot[1 + \alpha(\theta - 20)]$, $R = k_R\cdot\rho_{20}\cdot L/S$ ($k_R = 1.125$ takes the pure-copper ρ_{20} to the cable standard's maximum conductor resistance; k enters with the pure ρ_{20})
Fault current	$I_f = C_{min}\cdot U_0/Z_s$, $C_{min} = 0.95$
Maximum loop impedance	$Z_{s,max} = C_{min}\cdot U_0/I_a$; I_a = 5, 10 or 20·I_n for MCB types B, C, D, or read from the time–current points at t_{req} (log-log)
Adiabatic conductor check	$S_{min} = \sqrt{(I^2t)/k}$, $k = \sqrt{[Q_c(\beta + 20)/\rho_{20}\cdot \ln(1 + (\theta_f - \theta_i)/(\beta + \theta_i))]}$ (derived); $I^2t = I_f^2\cdot t_d$ from 0.1 s; below 0.1 s the device's let-through I^2t at the maximum prospective fault, which bounds the protective and the line conductor at the origin where the earth-fault current is largest (reg. 434.5.2, 543.1.3); without it neither is checked at the origin, the sheet fails 'Conductors against I...
Protective device check	the device chosen here must be the one carried from the cable step ($device_{code}$ 1 = MCB B, 2 = C, 3 = D, 4 = fuse or MCCB), else the sheet is blocked
Fault level (simplified)	$Z_T = u_k\cdot U_n^2/S_r$, $I_{k3} = c\cdot U_n/(\sqrt{3}\cdot Z_T)$, $U_n = \sqrt{3}\cdot U_0$

Basis: Automatic disconnection in TN systems: times and the loop impedance condition[1](reg. 411.3.2.2, 411.3.2.3, 411.4.4); Protective conductor and fault current adiabatic checks[1](reg. 543.1.3, 434.5.2); Protection by automatic disconnection of supply[2](cl. 6.3.3); Voltage factor c[3](cl. 5.3).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
U_0	Nominal voltage to earth	230	V	spec: Supply characteristics	50 ... 1000 V
Z_e	External earth fault loop impedance	0.8	ohm	spec: DNO declared maximum for TN-S	0.01 ... 3 ohm
C_{min}	Voltage factor C_{min}	0.95		assumption: Minimum voltage factor.	0.9 ... 1

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
c_{max}	Voltage factor c for the maximum fault	1.1		assumption: Maximum voltage factor.	1 … 1.1
L	Circuit length	40	m	drawing: E-LAY-002 rev B	0.1 … 1000 m
S_{line}	Line conductor cross-section	4	mm²	drawing: E-SCH-002 rev B	0.5 … 630 mm²
S_{cpc}	Protective conductor cross-section	2.5	mm²	drawing: E-SCH-002 rev B	0.5 … 630 mm²
θ	Conductor temperature for Z_s	70	°C	assumption: Thermoplastic insulation at full load.	20 … 90 °C
ρ_{20}	Resistivity at 20 °C (used for k, and for $R_1 + R_2$ through k_R)	0.000017241	ohm·mm	assumption: Annealed copper resistivity.	0.000001 … 0.0001 ohm·mm
k_R	Conductor resistance factor on ρ_{20} for $R_1 + R_2$ (1 = the resistivity itself; the cable standard's maximum conductor resistances are 5 to 12 % above pure copper, so the default is 1.125 and Z_s is not understated; k is not changed by it)	1.125		assumption: Cable standard maximum conductor resistance over the pure-copper resistivity (5 to 12 % higher); 1.125 covers the sizes used.	1 … 1.5
α	Temperature coefficient of resistance	0.00393	1/K	assumption: Copper temperature coefficient.	0.001 … 0.01 1/K
Q_c	Volumetric heat capacity	0.00345	J/(K·mm³)	assumption: Copper volumetric heat capacity.	0.001 … 0.01 J/(K·mm³)
β	Reciprocal temperature coefficient β	234.5	K	assumption: Copper reciprocal temperature coefficient.	100 … 400 K
θ_i	Initial temperature for the adiabatic check	70	°C	assumption: Thermoplastic operating temperature.	20 … 250 °C
θ_f	Final temperature for the adiabatic check	160	°C	assumption: Thermoplastic limiting temperature.	100 … 400 °C
device	Protective device	MCB type C (I_α = 10 I_n)		drawing: E-SCH-002 rev B	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
I_n	Rated current	16	A	drawing: E-SCH-002 rev B	1 … 1600 A
t_{req}	Required disconnection time	0.4	s	spec: Final circuit up to 32 A, TN system	0.1 … 5 s
$I2t_{let}$	Device let-through energy I^2t at the maximum prospective fault (disconnection below 0.1 s)	1500	$A^2·s$	assumption: Read from the MCB let-through chart at the fault current.	1 … 1000000000 $A^2·s$

3. Assumptions

A1. TN system, a single earth fault of negligible impedance; no motor contribution, arc fault or IT system. (Scope of the check.)
A2. Line and protective conductors are the same material at the same temperature; R = ρ_{20}·L/S corrected by α. (Conductor data from a material record.)
A3. k is derived from the material constants and labelled derived; it is not read from a licensed table. (Derived constants, not read from a table.)
A4. Fuse and MCCB characteristics are points the user reads off the manufacturer's curve (log-log interpolation; no extrapolation). (No time–current table ships.)
A5. Optional field-test value 0.8 × $Z_{s,max}$ is a rule of thumb for measurements at ambient temperature. (Labelled on the sheet.)
A6. The line conductor check (reg. 434.5.2) needs the device's let-through energy at the maximum prospective fault; the end-of-line earth fault is the minimum fault and does not bound it, so the check is made only when $I2t_{let}$ is entered or the disconnection time is 0.1 s or more. (Stated limitation: the end-of-line earth fault is the minimum fault, not the maximum.)
A7. C_{min} (Voltage factor C_{min}) is taken as 0.95. (Minimum voltage factor.)
A8. c_{max} (Voltage factor c for the maximum fault) is taken as 1.1. (Maximum voltage factor.)
A9. θ (Conductor temperature for Z_s) is taken as 70 degC. (Thermoplastic insulation at full load.)
A10. ρ_{20} (Resistivity at 20 °C (used for k, and for $R_1 + R_2$ through k_R)) is taken as 0.000017241 ohm*mm. (Annealed copper resistivity.)
A11. k_R (Conductor resistance factor on ρ_{20} for $R_1 + R_2$ (1 = the resistivity itself; the cable standard's maximum conductor resistances are 5 to 12 % above pure copper, so the default is 1.125 and Z_s is not understated; k is not changed by it)) is taken as 1.125. (Cable standard maximum conductor resistance over the pure-copper resistivity (5 to 12 % higher); 1.125 covers the sizes used.)
A12. α (Temperature coefficient of resistance) is taken as 0.00393 1/K. (Copper temperature coefficient.)
A13. Q_c (Volumetric heat capacity) is taken as 0.00345 J/(K*mm^3). (Copper volumetric heat capacity.)
A14. β (Reciprocal temperature coefficient β) is taken as 234.5 K. (Copper reciprocal temperature coefficient.)
A15. θ_i (Initial temperature for the adiabatic check) is taken as 70 degC. (Thermoplastic operating temperature.)
A16. θ_f (Final temperature for the adiabatic check) is taken as 160 degC. (Thermoplastic limiting temperature.)
A17. $I2t_{let}$ (Device let-through energy I^2t at the maximum prospective fault (disconnection below 0.1 s)) is taken as 1500 A^2*s. (Read from the MCB let-through chart at the fault current.)

4. Working

[19] $R_{12} = \rho_{20} * k_R * L * (1 / S_{line} + 1 / S_{cpc}) * (1 + \alpha * (\theta - 20\ degC))$
= 0.000017241 ohm·mm × 1.125 × 40 m × (1 / 4 mm² + 1 / 2.5 mm²) × (1 + 0.00393 1/K × (70 degC
degC))

= 0.60339 ohm

$R_1 + R_2$ at the conductor temperature

[20] $Z_s = Z_e + R_{12}$ [1](reg. 411.4.4)

= 0.8 ohm + 0.60339 ohm

= 1.4034 ohm

Earth fault loop impedance

[21] $I_f = C_{min} * U_0 / Z_s$

= 0.95 × 230 V / 1.4034 ohm

= 155.69 A

Earth fault current

[22] $I_a = 10 * I_n$ [2](60898-1)

= 10 × 16 A

= 160 A

Instantaneous tripping current (10 I_n)

[23] $Z_{s,max} = C_{min} * U_0 / I_a$ [1](reg. 411.4.4)

= 0.95 × 230 V / 160 A

= 1.3656 ohm

Maximum earth fault loop impedance

[24] $u_{Zs} = Z_s / Z_{s,max}$

= 1.4034 ohm / 1.3656 ohm

= 1.0277

Loop impedance utilisation

[25] $Z_{test} = 0.8 * Z_{s,max}$

= 0.8 × 1.3656 ohm

= 1.0925 ohm

Field-test maximum at ambient (0.8 × $Z_{s,max}$, rule of thumb)

5. Results

SYMBOL	VALUE	DESCRIPTION
R_{12}	0.60339 ohm	$R_1 + R_2$
Z_s	1.4034 ohm	Loop impedance
I_f	155.69 A	Earth fault current
I_a	160 A	Disconnection current
$Z_{s,max}$	1.3656 ohm	Maximum loop impedance
u_{Zs}	1.0277	Loop impedance utilisation
Z_{test}	1.0925 ohm	Field-test maximum (rule of thumb)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Disconnection achieved	I_f below the instantaneous tripping current	disconnection within t_{req}	–	[1](reg. 411.3.2.2)
FAIL	Loop impedance within maximum (disconnection in time)	u_{Zs} = 1.0277	$u_{Zs} \le 1$	1.03	[1](reg. 411.4.4)

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: ρ_{20}, k_R, α, Q_c, β, θ_i, θ_f (Conductor material constants); $I2t_{let}$ (Device time–current points). Enter each as one so its source, edition and licence travel with it.
WARNING — adiabatic: I_f: the device does not disconnect within the data given, so the fault energy and the adiabatic check cannot be stated; reduce the loop impedance or choose a faster device.

8. Verification

```
Template elec.fault-protection@1.3.1 Content hash 38d73f5c838ae5dd Engine calc-engine/1.0.0
Known-answer cases passed 6/6 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. British Standards Institution; Institution of Engineering and Technology. Requirements for Electrical Installations: IET Wiring Regulations. BS 7671. IET.
2. BS EN 60204-1:2018.
3. IEC 60909-0:2016.